Derivatives of Exponential Functions

Exam code: 9MA0

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Help me understand

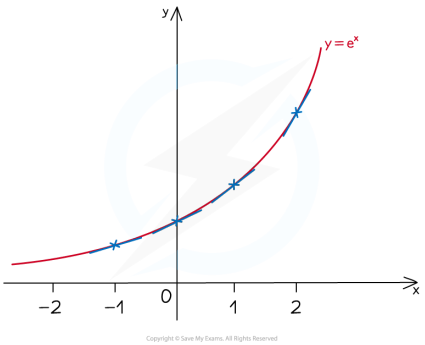
Derivatives of exponential functions

What is the derivative of e^x?

- $y = e^x$ has the particular property that

$$\frac{dy}{dx} = e^x$$

- i.e. for *every* real number **x**, the gradient of $y = e^x$ is also equal to e^x (see Derivatives of Exponential Functions)



Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

0	1	1
1	2.7182...	2.7182...
2	7.3890...	7.3890...

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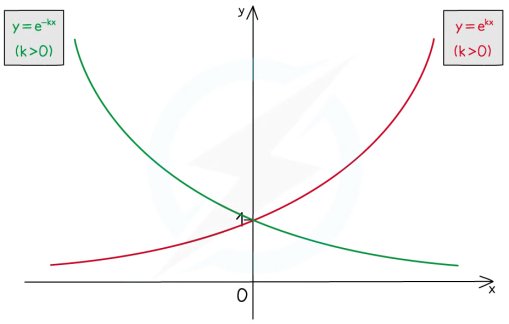
- **e ≈ 2.718** (see "e")
- Recall that the derivative is the gradient function for a curve (see First Principles Differentiation)

What is the derivative of y = e^{kx} and y=e^{-kx}?



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- The derivative of **y = e^{kx}** is
  - $\frac{dy}{dx} = k e^{kx}$
- The derivative of **y = e^{-kx}** is
  - $\frac{dy}{dx} = -k e^{-kx}$

Examiner Tips and Tricks

- Remember that (like **π**) **e** is a **number**.
- Exam questions can ask for answers to be given as exact values in terms of **e** (see the Worked Example below).

Worked Example

of the following functions ...

(a) $y = e^{x}$

(b) $y = e^{5x}$

(c) $y = e^{-x}$

(d) $y = 3e^{2x}$

(e) $y = 5e^{-4x}$

a) $y = e^{x}$ $\quad \frac{dy}{dx} = e^{x}$

$\frac{dy}{dx} = y$

AT $x=3$, $\quad \frac{dy}{dx} = e^{3}$

$\frac{dy}{dx} = e^{3}$

FINAL ANSWER IN TERMS OF e

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b) $y = e^{5x}$ $\quad \frac{dy}{dx} = 5e^{5x}$

$\frac{dy}{dx} = ke^{kx}$

AT $x=3$, $\quad \frac{dy}{dx} = 5e^{5\cdot 3}$

$\frac{dy}{dx} = 5e^{15}$

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